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Module 2:

Ecosystem & Biodiversity

Dr. Saurav Prasad

EVS- An Insight

Key Environmental Problems and Sustainability

Global environmental issues, socio-environmental interaction; impact of consumerism and global economy; Sustainability: Fundamentals, science behind sustainability, challenges, potential; role of technology, policies.

Ecosystem and Biodiversity

Ecosystem: functions and services, Nutrient, phosphorous, carbon, nitrogen, cycles; Effect of human activities on these cycles. Threats to biodiversity: conservation strategies, sustainable solutions.

Environmental pollution

Pollution sources, effects and mitigation, conservation. Water footprint and virtual water. Sustainable solutions. Environmental Protection Act – Air, water, forest and wild life. Interactional protocols and agreements. Resource recycling and sustainability.

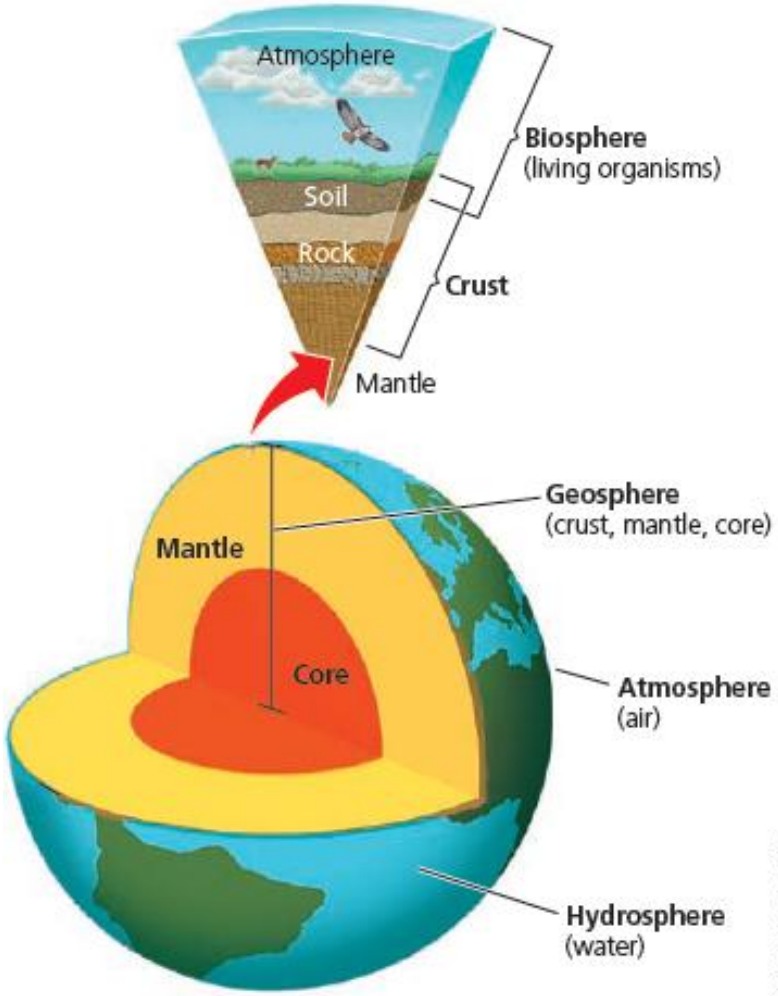
Human Population and Environment

Impact of population age structure, human population growth, Sustaining human societies: Economics, environment, policies and education.

Energy Sources and Sustainability

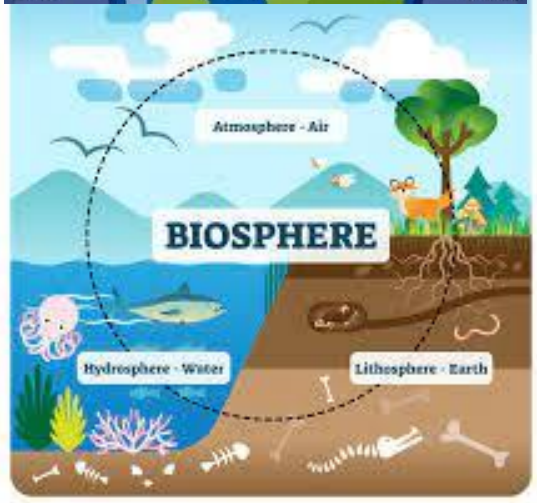
Conventional sources, non-conventional sources, natural resources and economics, clean coal technology, sequestration methods, energy conservation and sustainability, policies, acts, examples.

Earth's Life-Support System:
Four Major Components





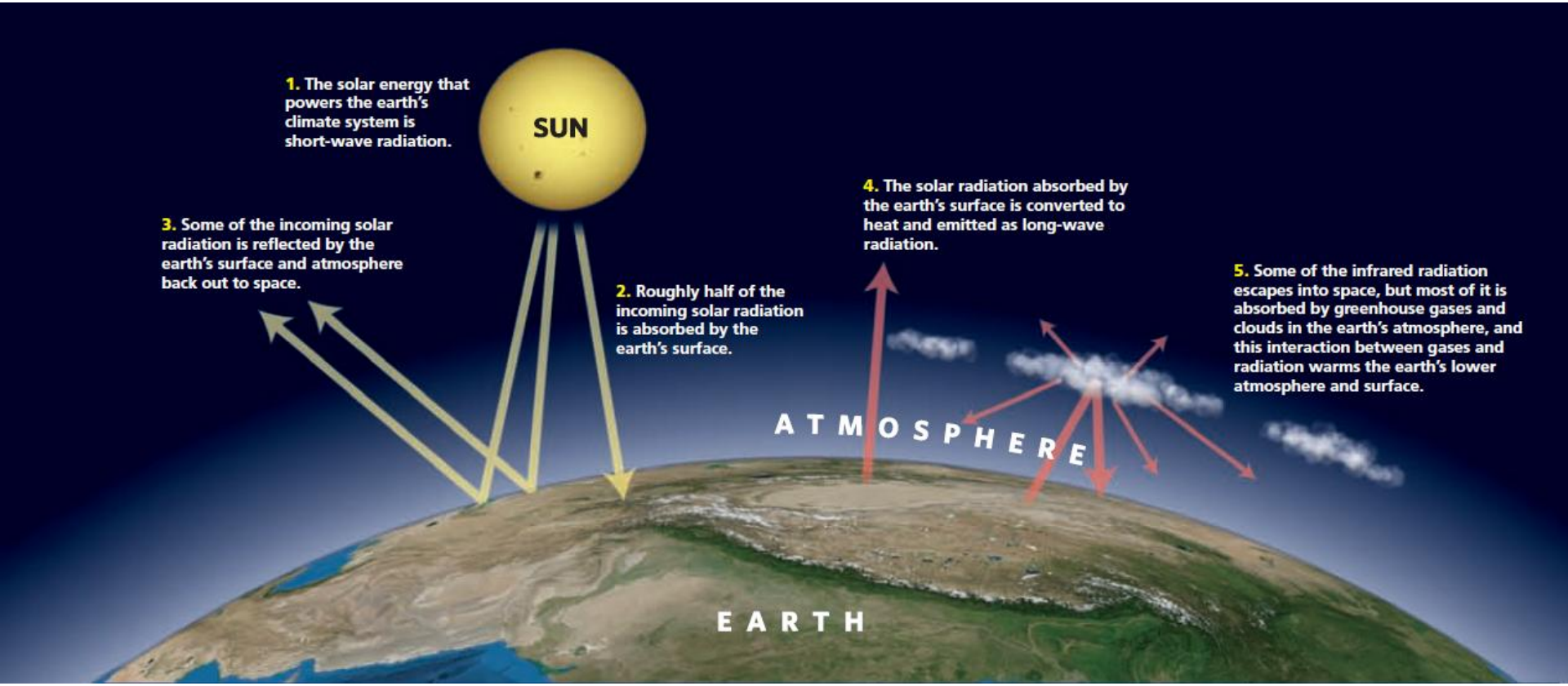




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Three Factors Sustain the Earth's Life!



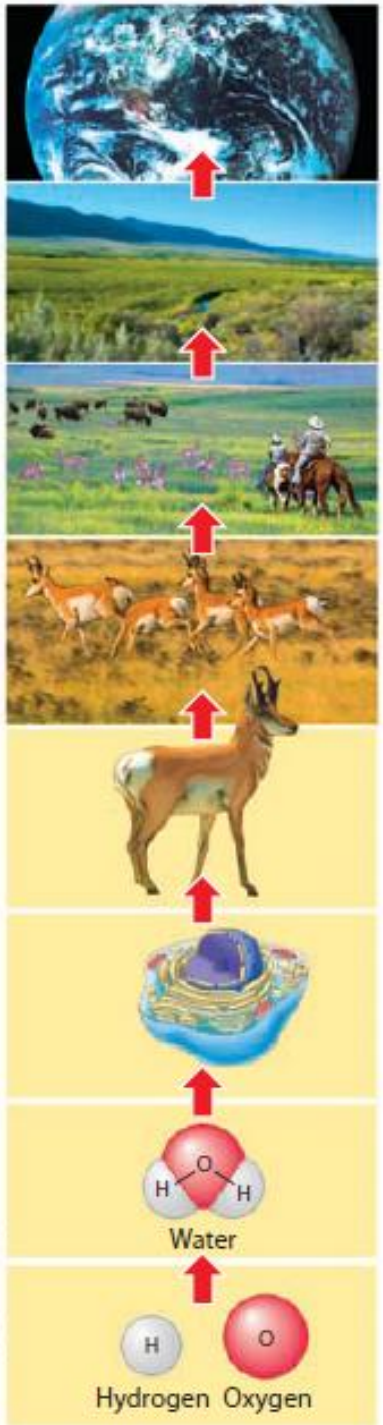
- ✓ One way flow of high quality energy
- ✓ Cycling of nutrients
- ✓ Gravity

Ecosystem

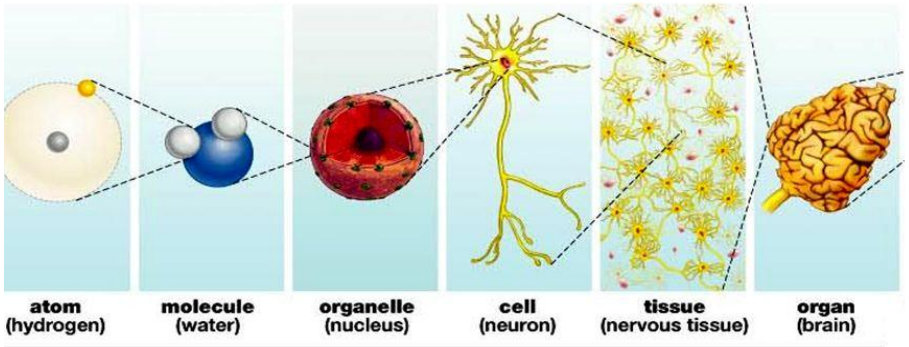
An ecosystem is a self-regulating group of biotic communities of species interacting with one another and with their non-living environment exchanging energy and matter.

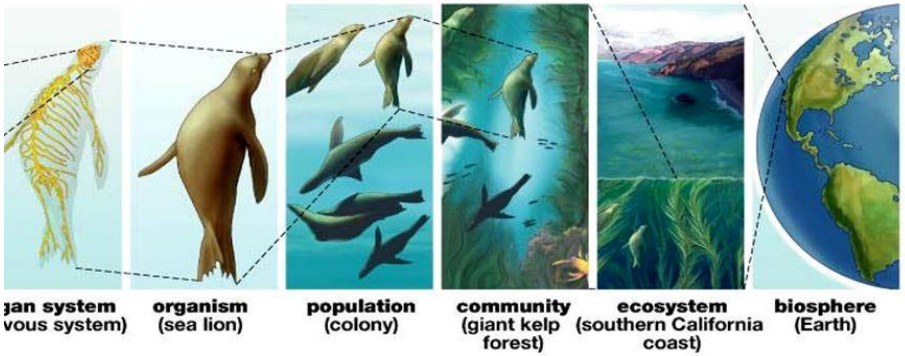


Organization of Matter in Nature!



Biosphere	Parts of the earth's air, water, and soil where life is found
Ecosystem	A community of different species interacting with one another and with their nonliving environment of matter and energy
Community	Populations of different species living in a particular place, and potentially interacting with each other
Population	A group of individuals of the same species living in a particular place
Organism	An individual living being
Cell	The fundamental structural and functional unit of life
Molecule	Chemical combination of two or more atoms of the same or different elements
Atom	Smallest unit of a chemical element that exhibits its chemical properties





Ecosystem

I: Natural: Terrestrial



Grassland



Forest



Desert

Natural: Aquatic



a) Marine: Oceans



b) Freshwater: Lakes

Lentic standing type, ponds, lakes

Lotic free-flowing rivers

II: Artificial or Manmade

Aquarium



Crop field



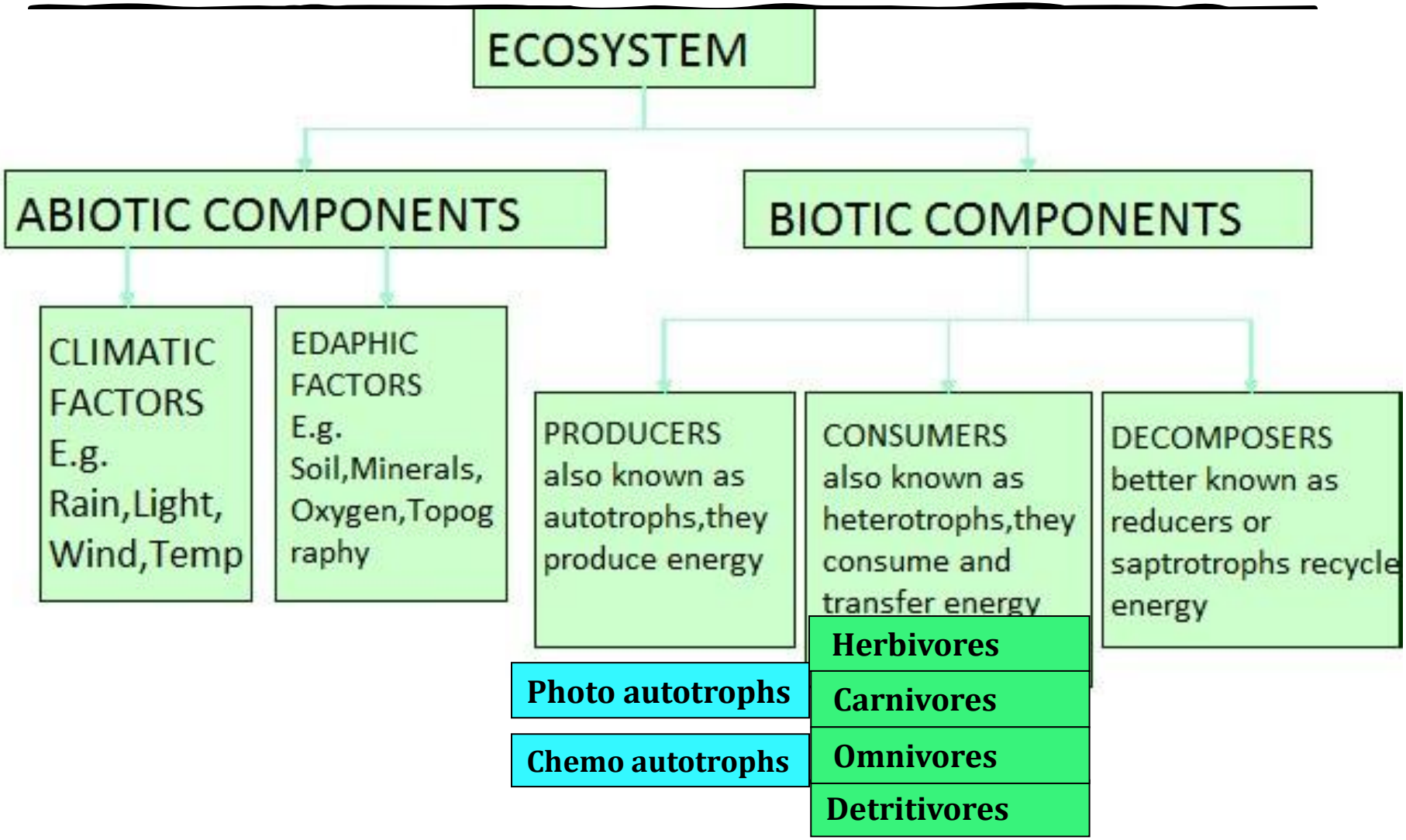
Structural Features of Ecosystems

Biotic factors
(living things)

Abiotic factors
(nonliving things)



Structure of Ecosystem



Biotic Structure

- Ecologists assign each organism in an ecosystem to a *feeding level,* or **trophic level**, depending on its source of nutrients.
- **Producers**, sometimes called **autotrophs** (self-feeders), make the nutrients they need from compounds and energy obtained from their environment.
- In a process called **photosynthesis**, plants capture solar energy and use it in combination with carbon dioxide and water to form organic molecules, which store the chemical energy that plants need.

$$6\ CO_2 + 6\ H_2O + \text{solar energy} \rightarrow C_6H_{12}O_6 + 6\ O_2$$

- The other organisms in an ecosystem are **consumers**, or **heterotrophs** ("other-feeders"), which cannot produce the nutrients they need through photosynthesis or other processes.
- They get their nutrients by feeding on other organisms (producers or other consumers) or their wastes and remains.

Biotic Structure: Producers

The plants, animals and microorganisms form the biotic component

Producers :



Photo autotrophs :

photo : light, auto: self, troph : food

Green plants which can synthesize their food by themselves by photosynthesis. They are also known as photo autotrophs.

Chemo-autotrophs (chemosynthetic) :

Chemo-autotrophs use inorganic energy sources such as hydrogen sulfide, elemental sulfur, ferrous iron, molecular hydrogen, and ammonia as their food



Example: chemo-autotrophic sulfur bacteria in the ocean depth convert H_2S and CO_2 into organic compound making use of the heat generated by the decay of radioactive elements present in the earth's core and released into the ocean's depths.

Biotic Structure: Consumers

All organisms which get their organic food by feeding up on other organisms are called consumers.

1. **Herbivores (Plant eaters) :** They feed directly on producers and hence also known as **primary consumers e.g., Rabbit, caterpillars**
2. **Carnivores (Meat eaters):** They feed on other consumers. For example if they feed on herbivores they are called **secondary consumers** (**e.g Frog**), but if they feed on other carnivores (**snake, big fish etc**) they are called **tertiary consumers**.
3. **Omnivores :** They feed on both animals and plants, **e.g. Humans, rat, fox, many birds**
4. **Detritivores :** They feed on the parts of dead organism, waste of living organisms, **e.g. Beetles, termites, ants, crabs, earthworms etc.**

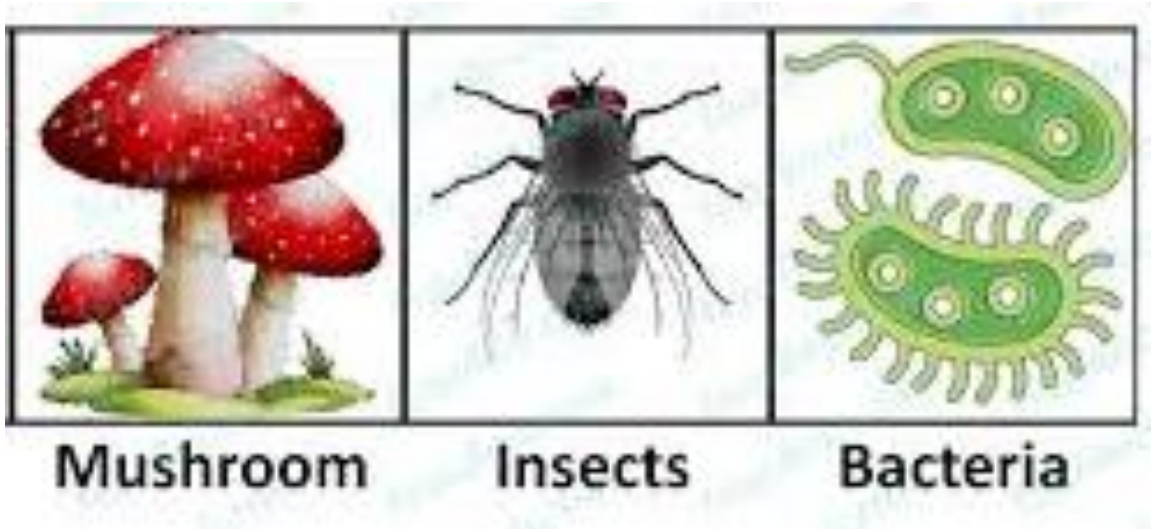






Biotic Structure: Decomposers

Derive nutrition by breaking down the complex organic molecules to simpler organic compounds and ultimately into inorganic nutrients.



This biotic structure prevails in all ecosystems. However, in some, it is the primary producers which dominate (e.g. forests, agroecosystems) and in some, decomposers predominate (e.g. deep ocean).

Ecosystem-Abiotic component

Physical factors

(a) Climatic factors

- Sunlight and shade
- Temperature
- Humidity
- Rainfall
- Wind

(b) Geographic factors

-Latitude

-Longitude

-Altitude

(c) Edaphic factors

-Soil type

-Soil moisture

-Soil reaction

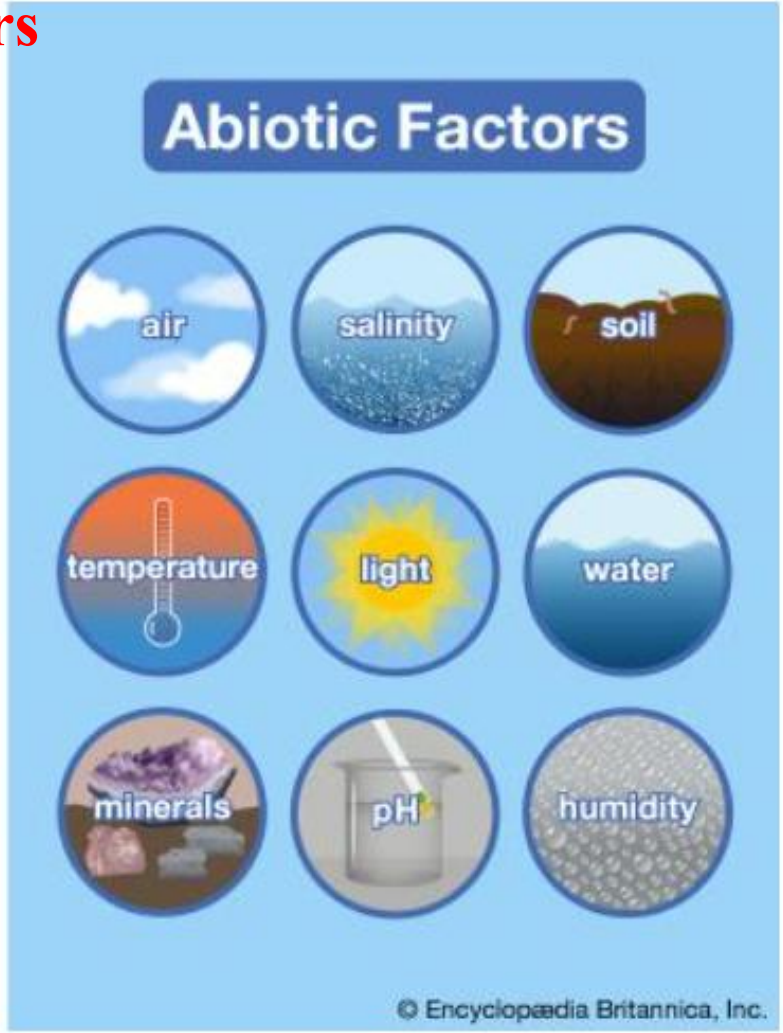
Chemical factors

(i) Essential nutrients such as C, N, P, $K, H_2, O_2, S,$

(ii) Trace nutrients

(iii) pollutants: level of toxic substances, salts causing salinity

(iv) Various organic substances present in soil and water



Importance of Ecosystem

1. **Biodiversity:** All the millions of species that exist on planet earth are sustained by their particular ecosystems.
2. **Evolution:** Organisms within an ecosystem have evolved to subsist within that ecosystem.
3. **Interconnectedness:** The organisms in any given ecosystem are usually highly interconnected. As such, it is important to understand that no species exists in total isolation.
4. **Self sustaining:** The delicate balance of organisms within an ecosystem helps to keep that ecosystem going.

   For example, removing one species of insect from a forest ecosystem can effect sustaining vast numbers of birds and pollinating vast numbers of flowers, trees and shrubs in the ecosystem.
5. **A reminder of the wild world:** Watching how ecosystems work in a perfect balance can remind humans of the fact that nature has its own rules and that we interfere with the workings of nature at great risk.
6. **Regulating the climate:** Many types of ecosystem protect the earth against extreme weather: forests provide barriers to floods and storms and also prevent the soil from being eroded by the rain
7. **Sources of food and fuel:** ROTI, KAPADA and MAKAN.

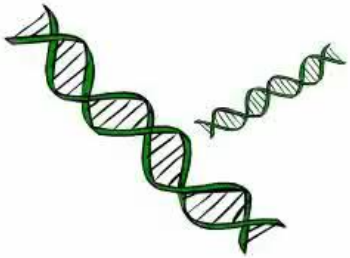
Activity 2

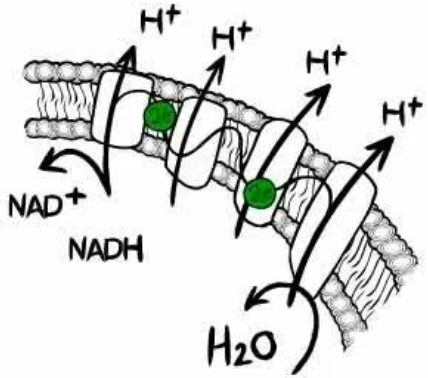
1. Which ecosystem do we live in?
2. Which is the largest ecosystem in the world?
3. What makes a good ecosystem?
4. Difference between Biome and Ecosystem?

Recap 1

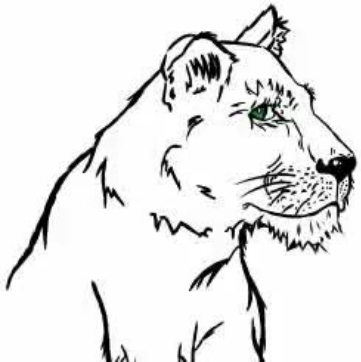
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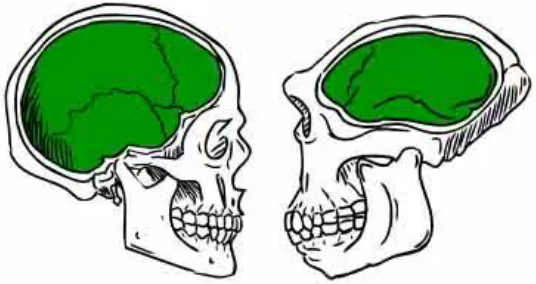




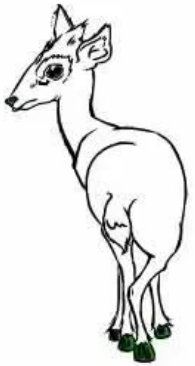


What is an ecosystem?









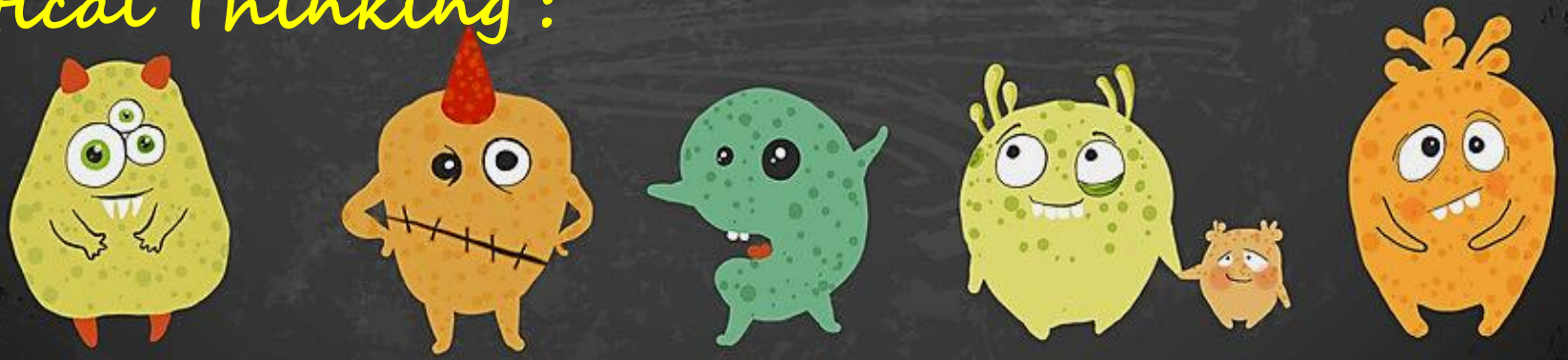
Recap 2

MANY OF THE WORLD'S MOST IMPORTANT ORGANISMS ARE INVISIBLE TO US!!





Critical Thinking :

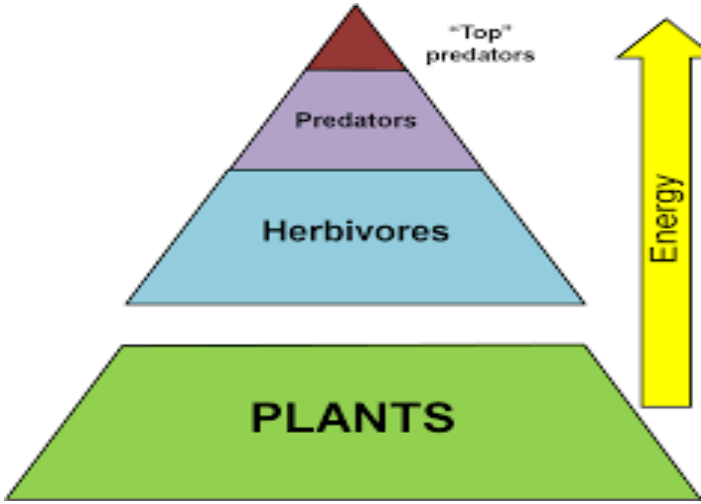


What are two advantages that microbes have over humans for thriving in the world?

What happens to energy in an ecosystem?

Trophic Structure

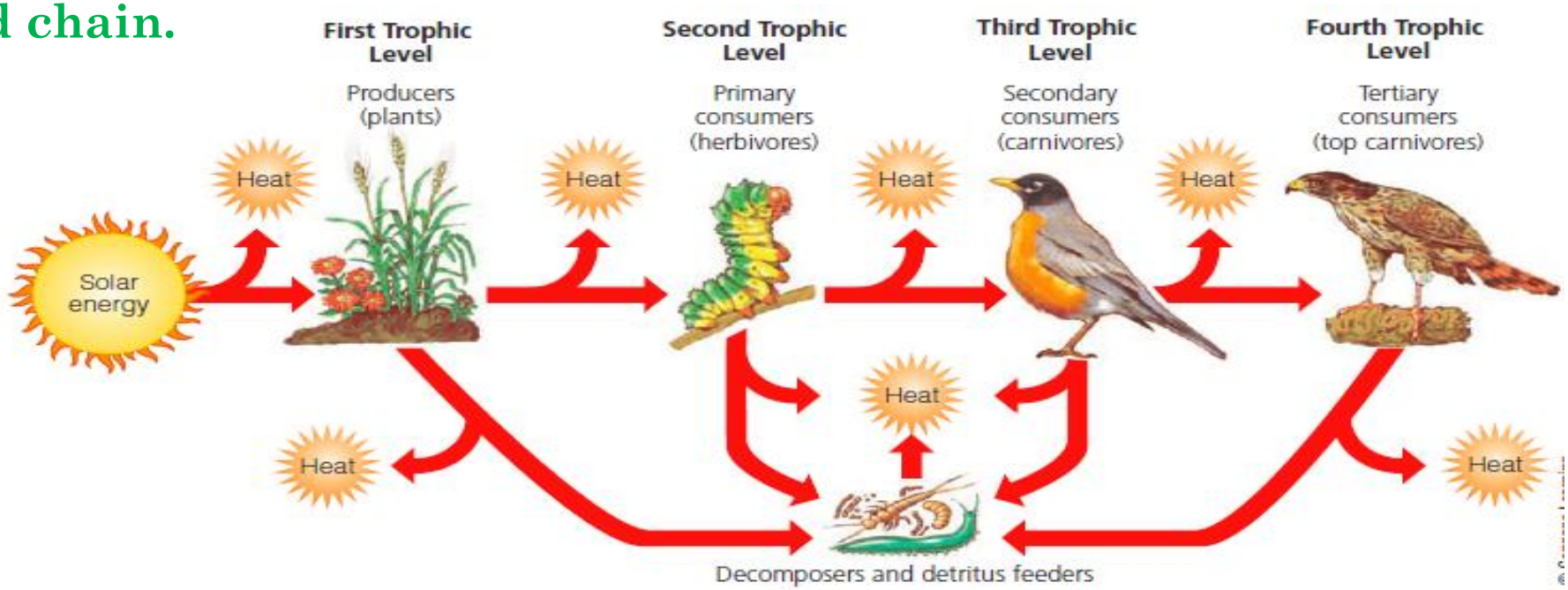
- The producers and consumers are arranged in the ecosystem in a definite manner and their interaction along with population size are expressed together as trophic structure.
- Each food level is known as trophic level.
- Amount of living matter at each trophic level at a given time is known as standing crop or standing biomass.



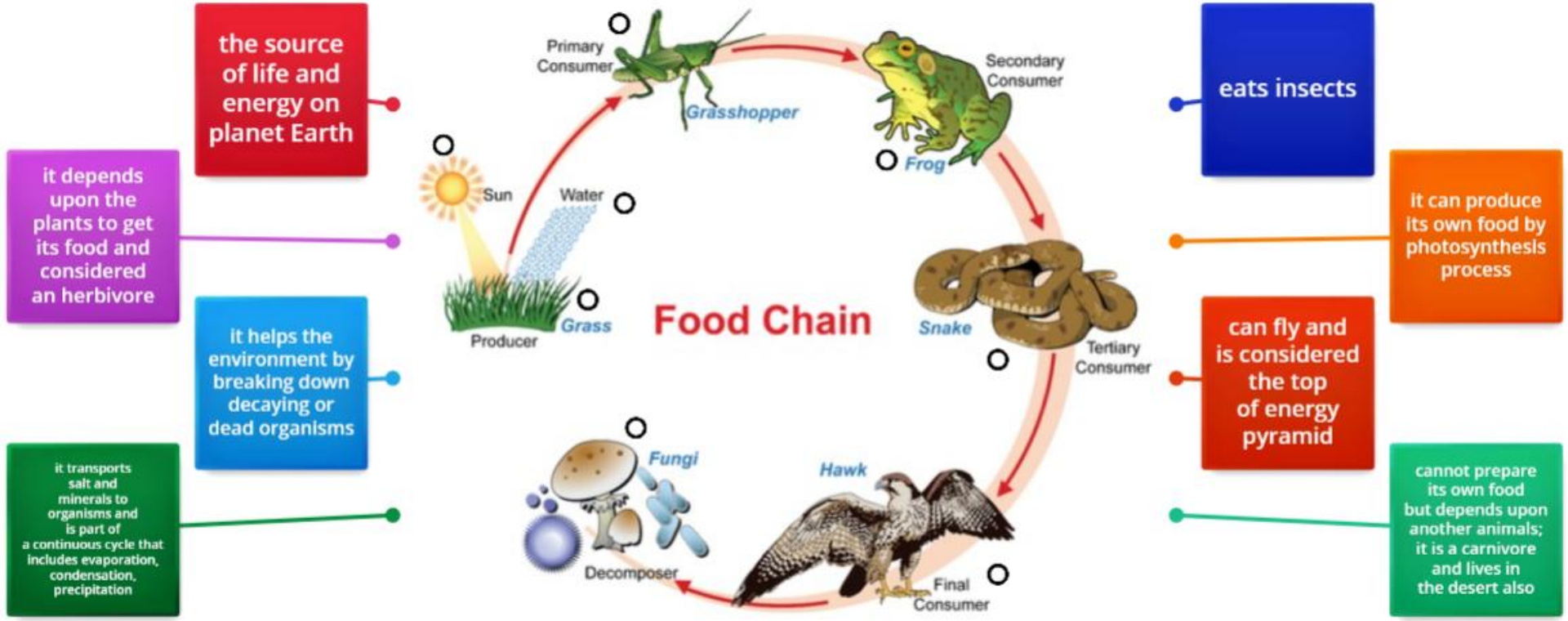
Food Chain (Trophic level)

The chemical energy stored as nutrients in the bodies and wastes of organisms **flows through ecosystems from one trophic (feeding) level to another.**

A sequence of organisms, each of which serves as a source of nutrients or energy for the next, is called a food chain OR the sequence of eating and being eaten in an ecosystems is known as food chain.



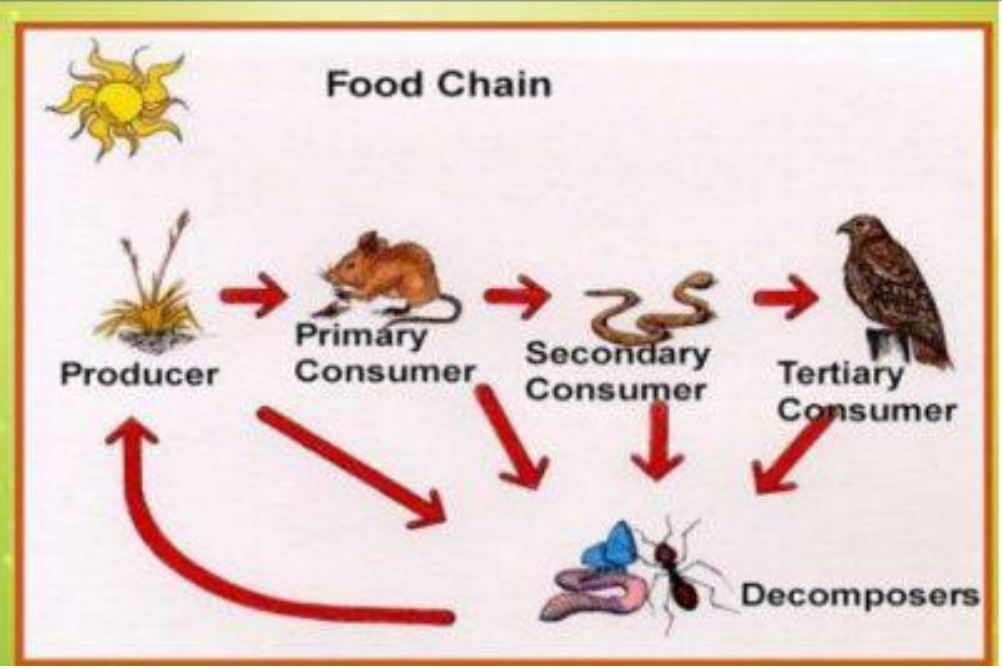
Play the game!

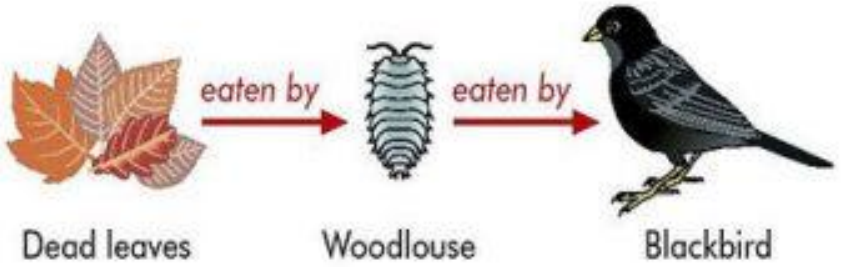


https://wordwall.net/resource/1636518/science/food-chain-diagram

Food Chain: 2 major types

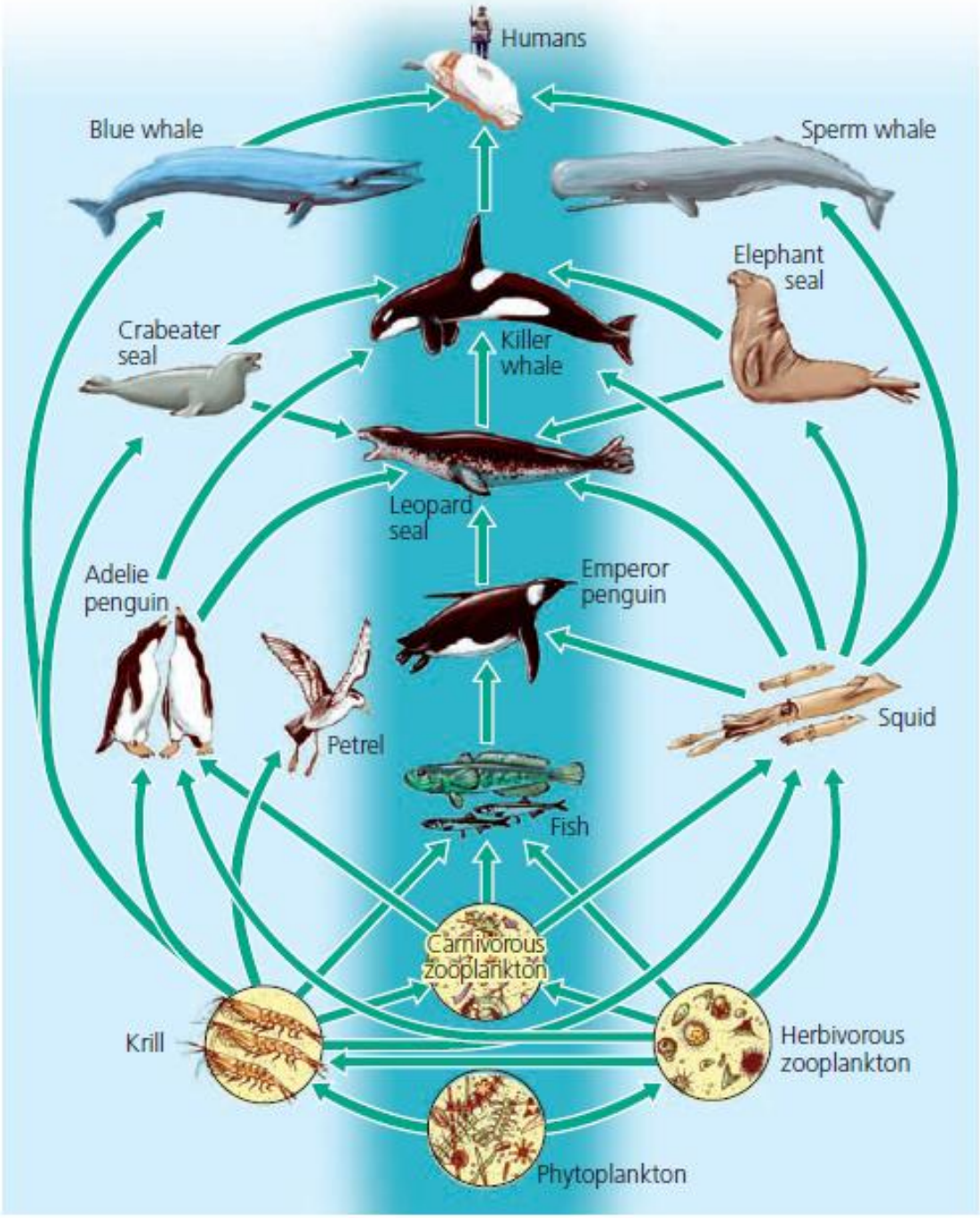
GRAZING FOOD CHAIN VS DETRITUS FOOD CHAIN





Food Web

- **In natural ecosystems, most consumers feed on more than one type of organism, and most organisms are eaten or decomposed by more than one type of consumer.**
- **Because of this, organisms in most ecosystems form a complex network of interconnected food chains called a food web.**
- **Food chains and food webs show how producers, consumers, and decomposers are connected to one another as energy flows through trophic levels in an ecosystem.**

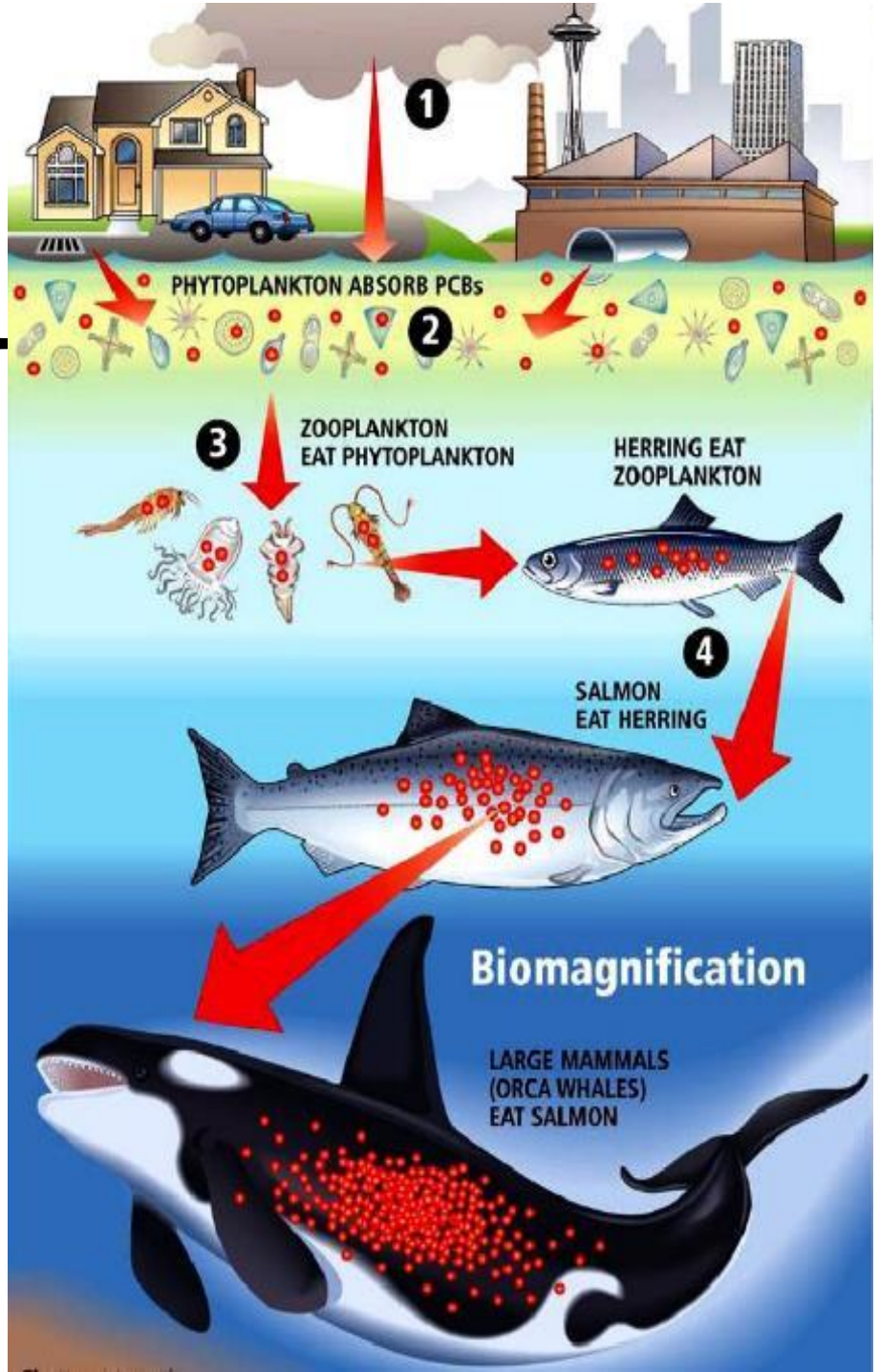


Dead Stuff: The Secret Ingredient in Our Food Chain



Significance of food chains and food webs

- Food webs **give greater stability to the ecosystem as compared to the food chain.**
- **Play significant role: nutrient and energy cycling**
- Maintain and regulate population size: **ecological balance**
- **Biological magnification:** the increase in concentration of harmful substances at higher levels of food chain



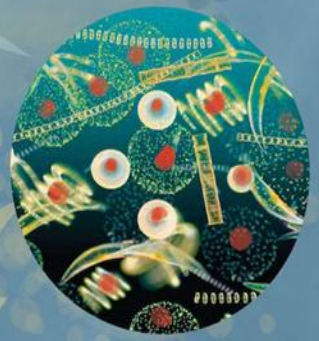
BIOMAGNIFICATION AND BIOACCUMULATION

How can pollutants have long-term effects on organisms?

Even when pollutants are not dangerous enough to kill animals outright, their presence can have lasting effects on food webs through **bioaccumulation** and **biomagnification**.

Toxins may increase in concentration as they are passed up the food chain, a process called **biomagnification**.

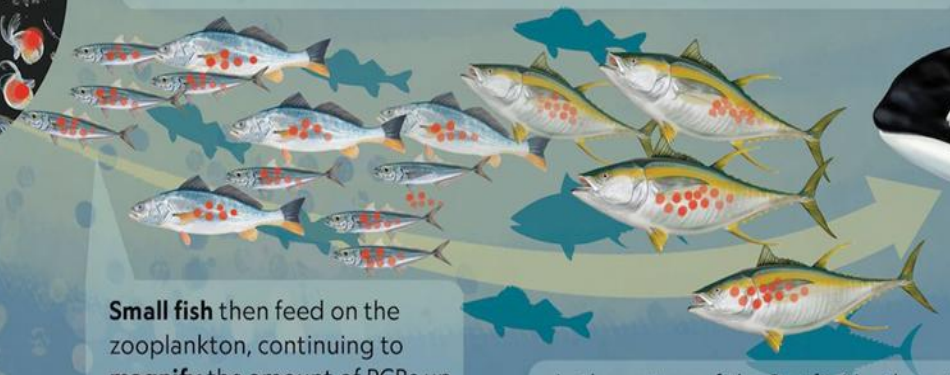
Pollutants such as **polychlorinated biphenyls (PCBs)** enter the ocean as industrial waste and are absorbed by microscopic **phytoplankton** at the bottom of the food chain.





PCBs

Even though phytoplankton absorb only a tiny amount, small creatures called **zooplankton** eat large quantities of the phytoplankton, taking in all the PCBs from what the phytoplankton eat.



Small fish then feed on the zooplankton, continuing to **magnify** the amount of PCBs up the food chain.



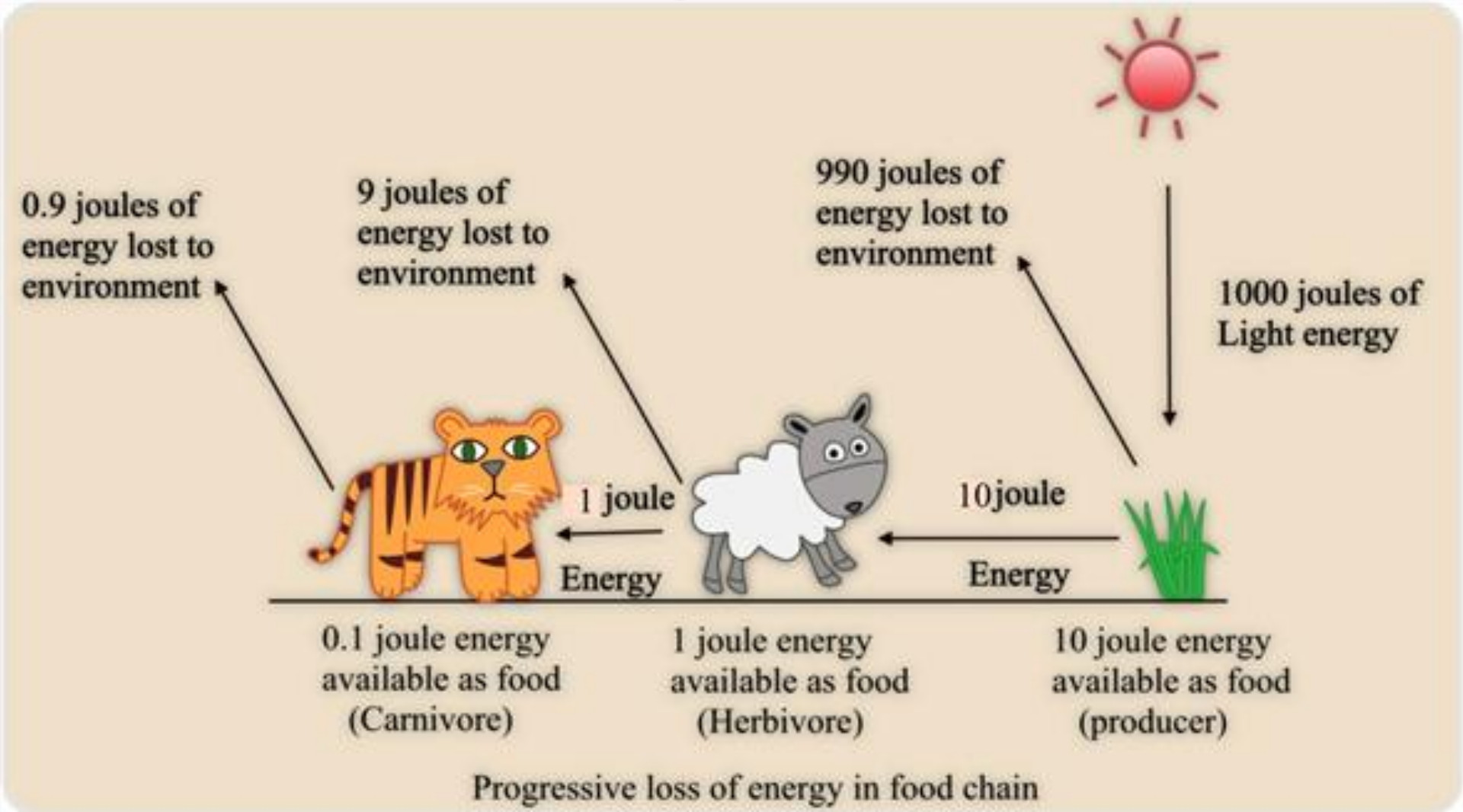
In the waters of the Pacific Northwest, **apex predators** like the killer whale (Orcinus orca) end up with the highest concentrations of toxins due to biomagnification.



Bioaccumulation occurs when pollutants build up in a single organism's body over time. Mercury, for example, is a pollutant that has entered waterways and lakes through industrial processes. Fish and shellfish absorb the mercury directly from their environment, and although they may only absorb small amounts at a time, the mercury can remain in the fish's body for months or even longer. This leads to the mercury building up, or **accumulating**, in the fish's body, posing a danger to any organism that eats the fish.

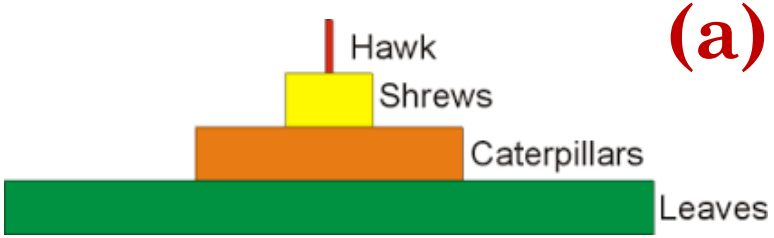
NATIONAL GEOGRAPHIC

Lindeman's 10% Law of Energy Flow

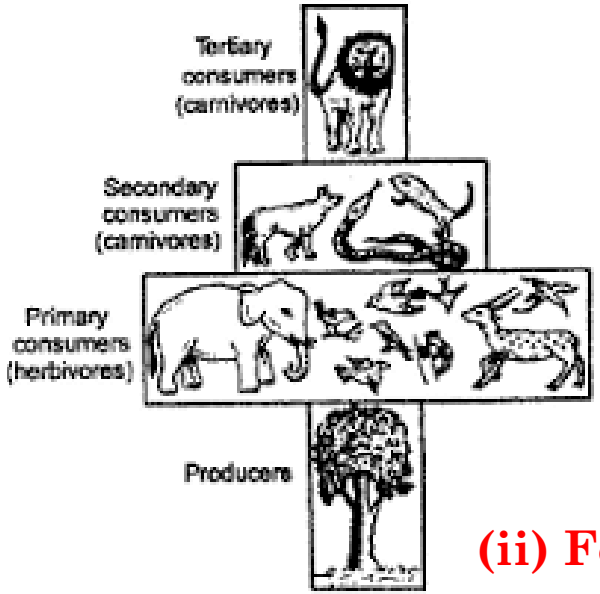


Ecological Pyramids

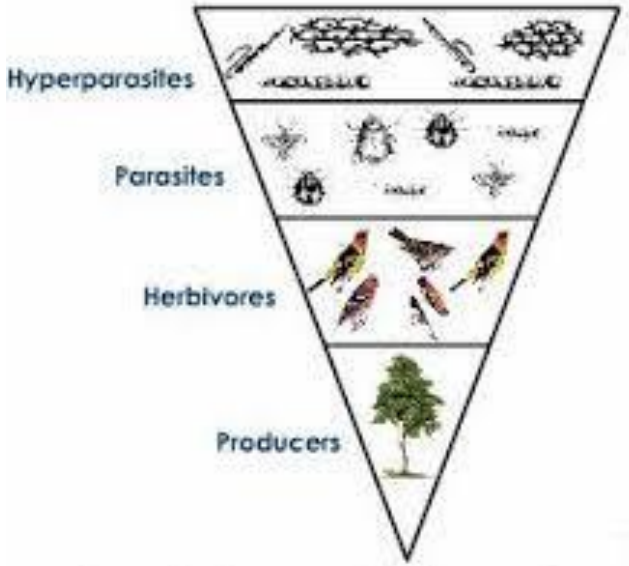
(a) Pyramid of numbers



(i) Grassland



(ii) Forest ecosystem



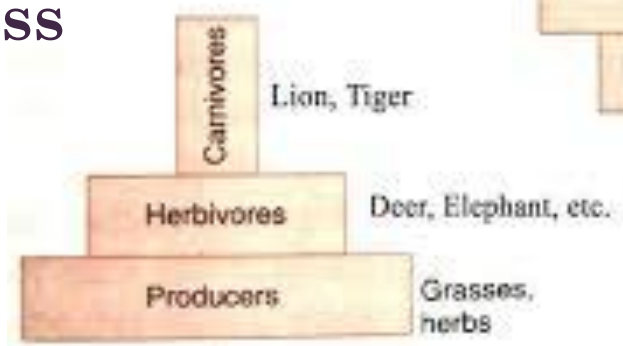
Inverted pyramid of number

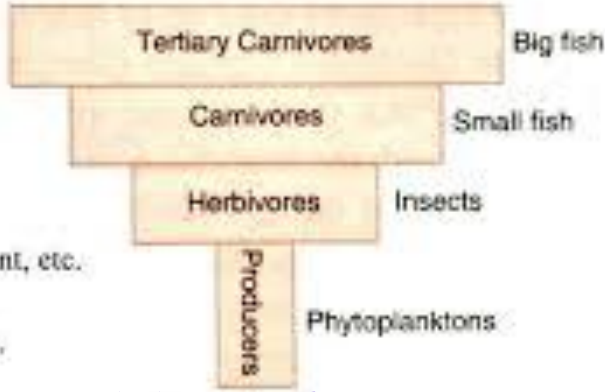
(iii) Parasitic food chain

Ecological Pyramids

(b) Pyramid of biomass

(i) Grassland ecosystem

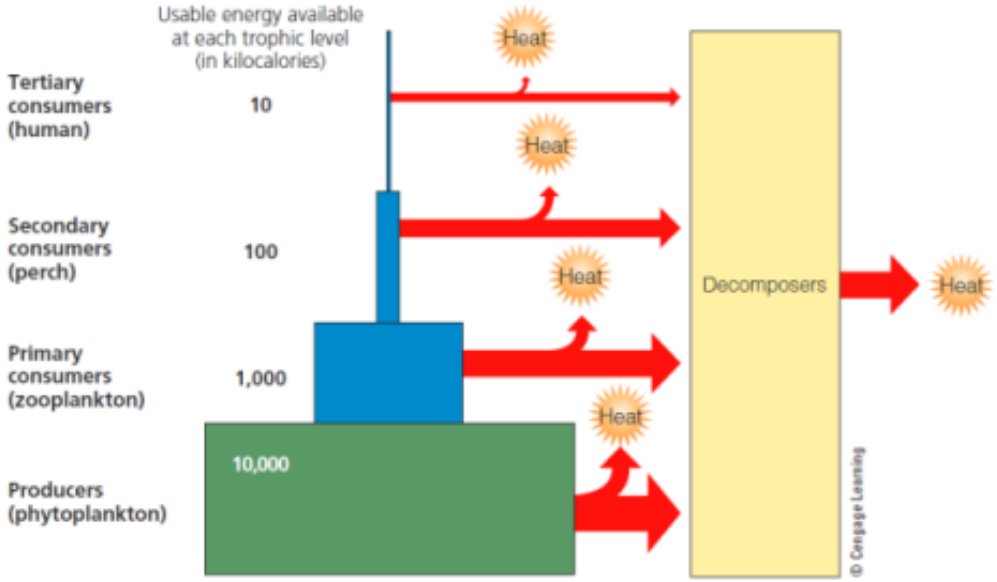




(ii) Pond ecosystem

(c) Pyramid of energy flow:

energy loss for a simple food chain, assuming a 90% energy loss with each transfer.

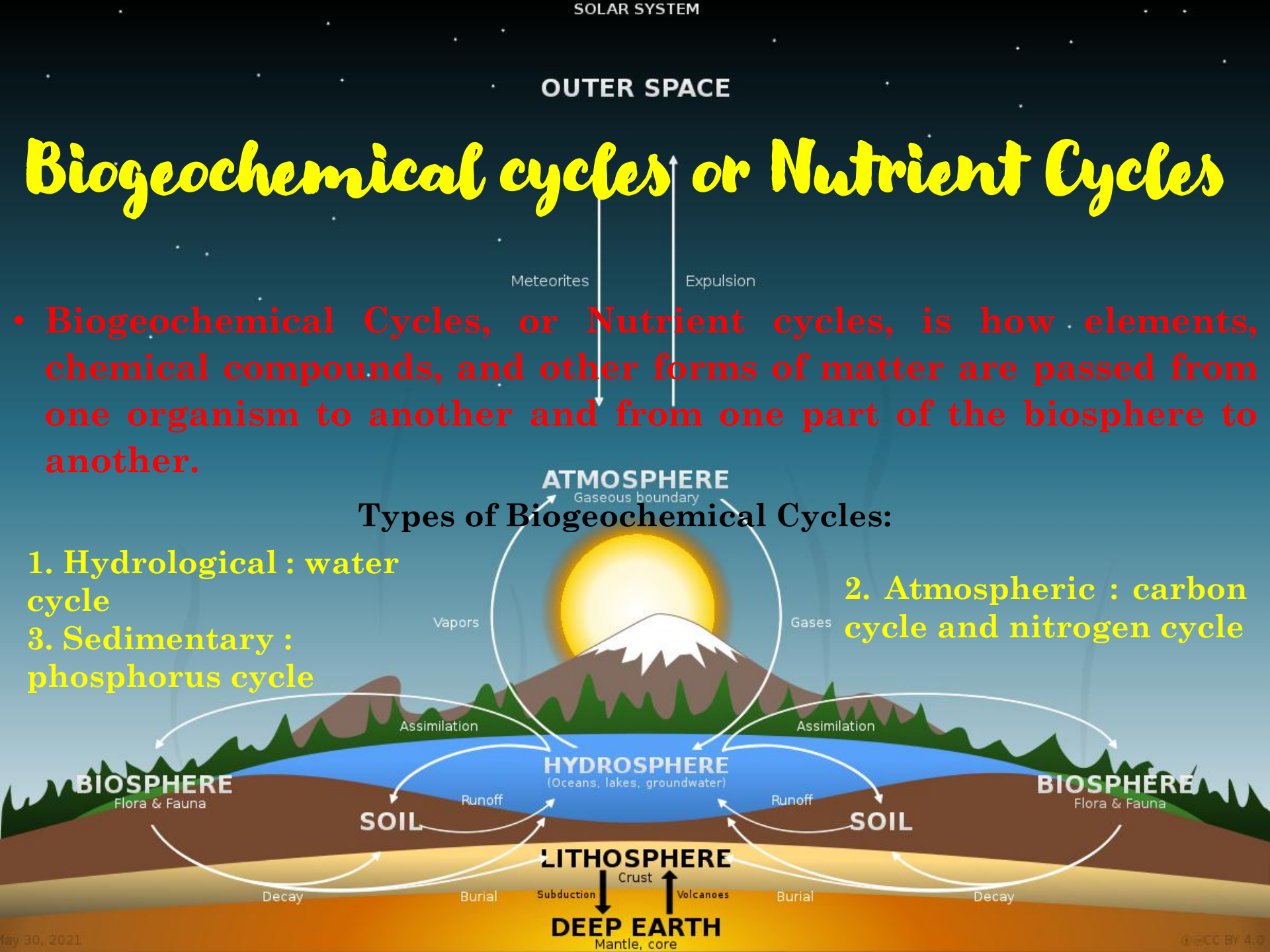


Biogeochemical cycles or Nutrient Cycles

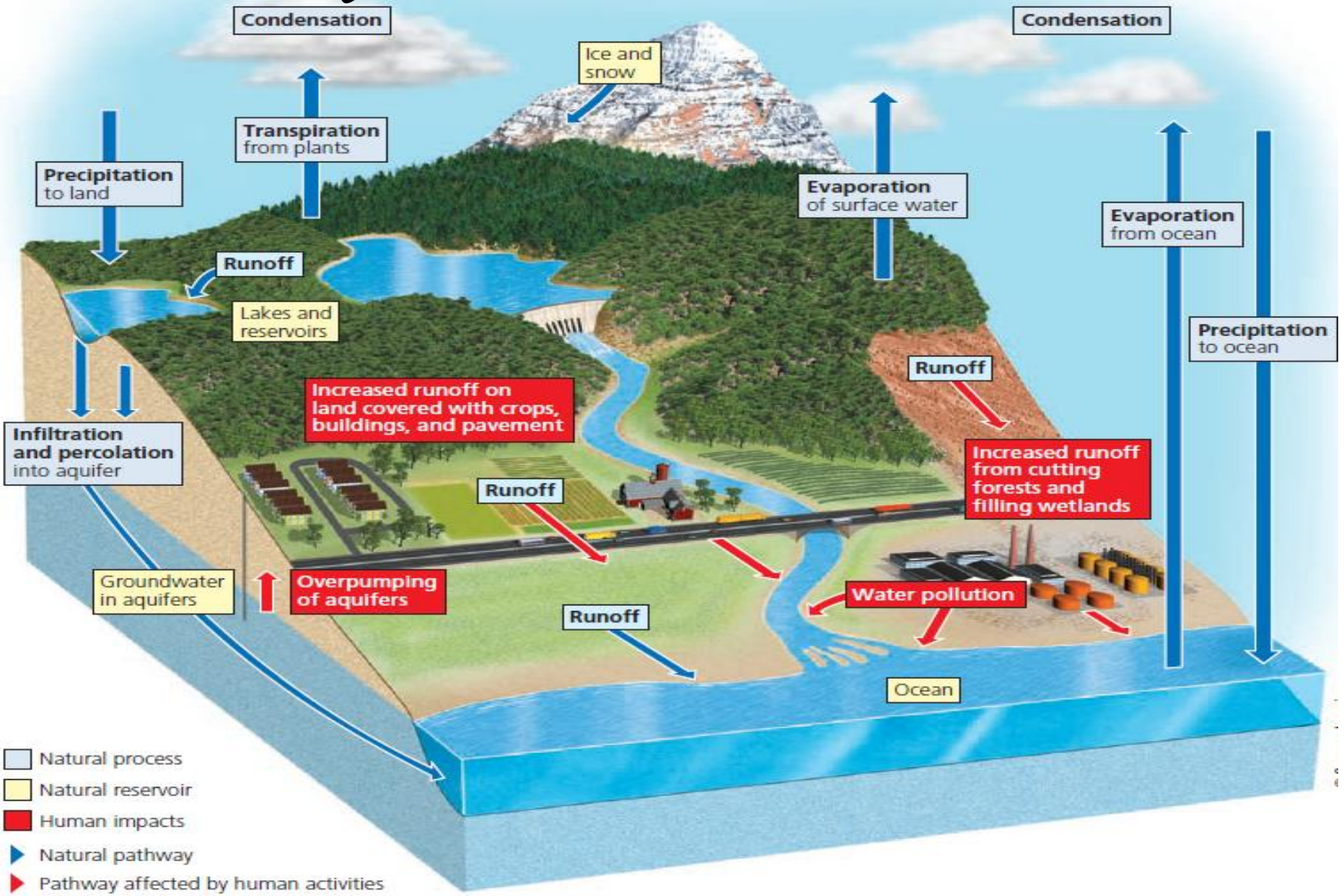
- Biogeochemical Cycles, or Nutrient cycles, is how elements, chemical compounds, and other forms of matter are passed from one organism to another.

Types of Biogeochemical Cycles:

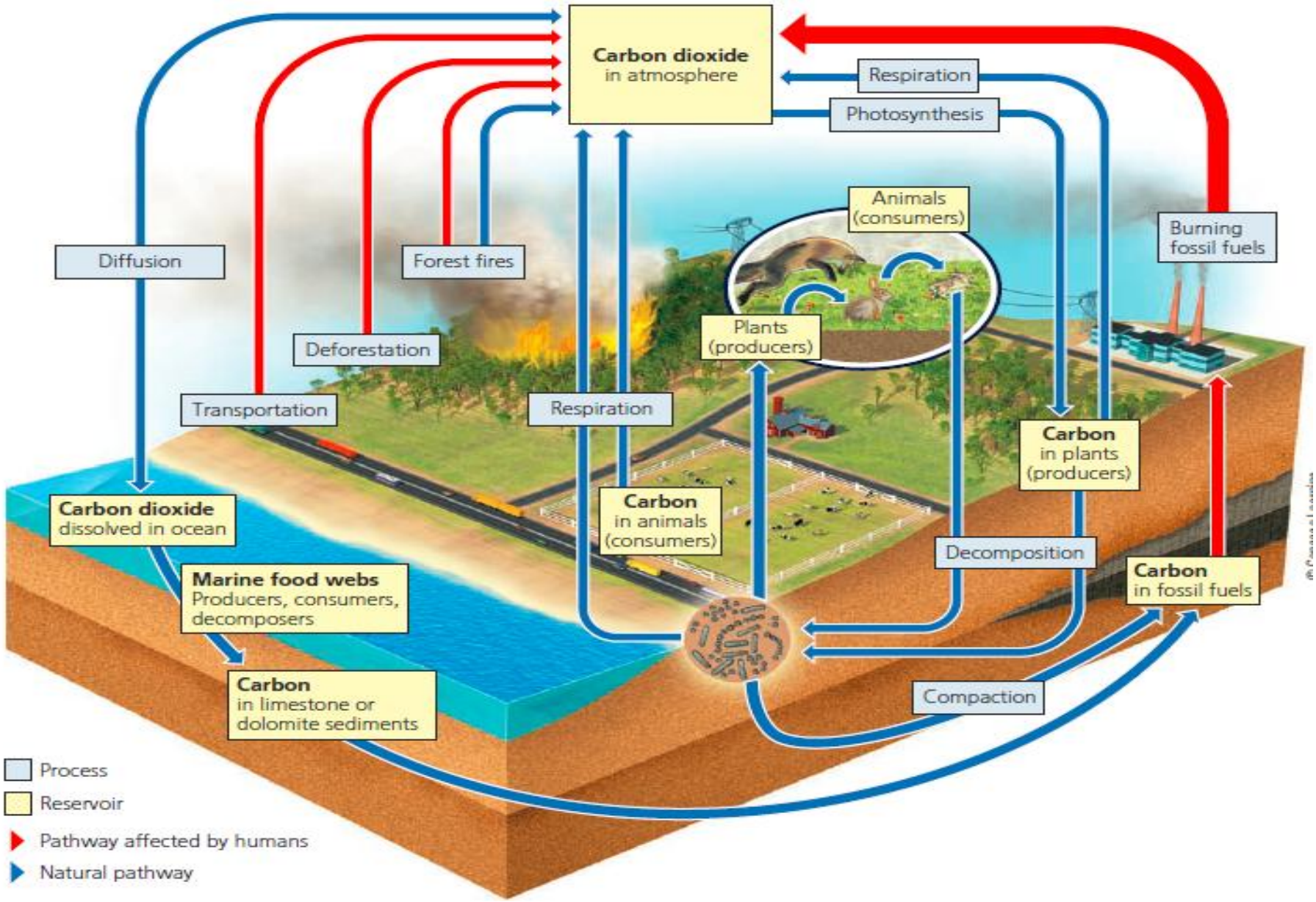
1. Hydrological : water cycle
2. Atmospheric : carbon cycle and nitrogen cycle
3. Sedimentary : phosphorus cycle



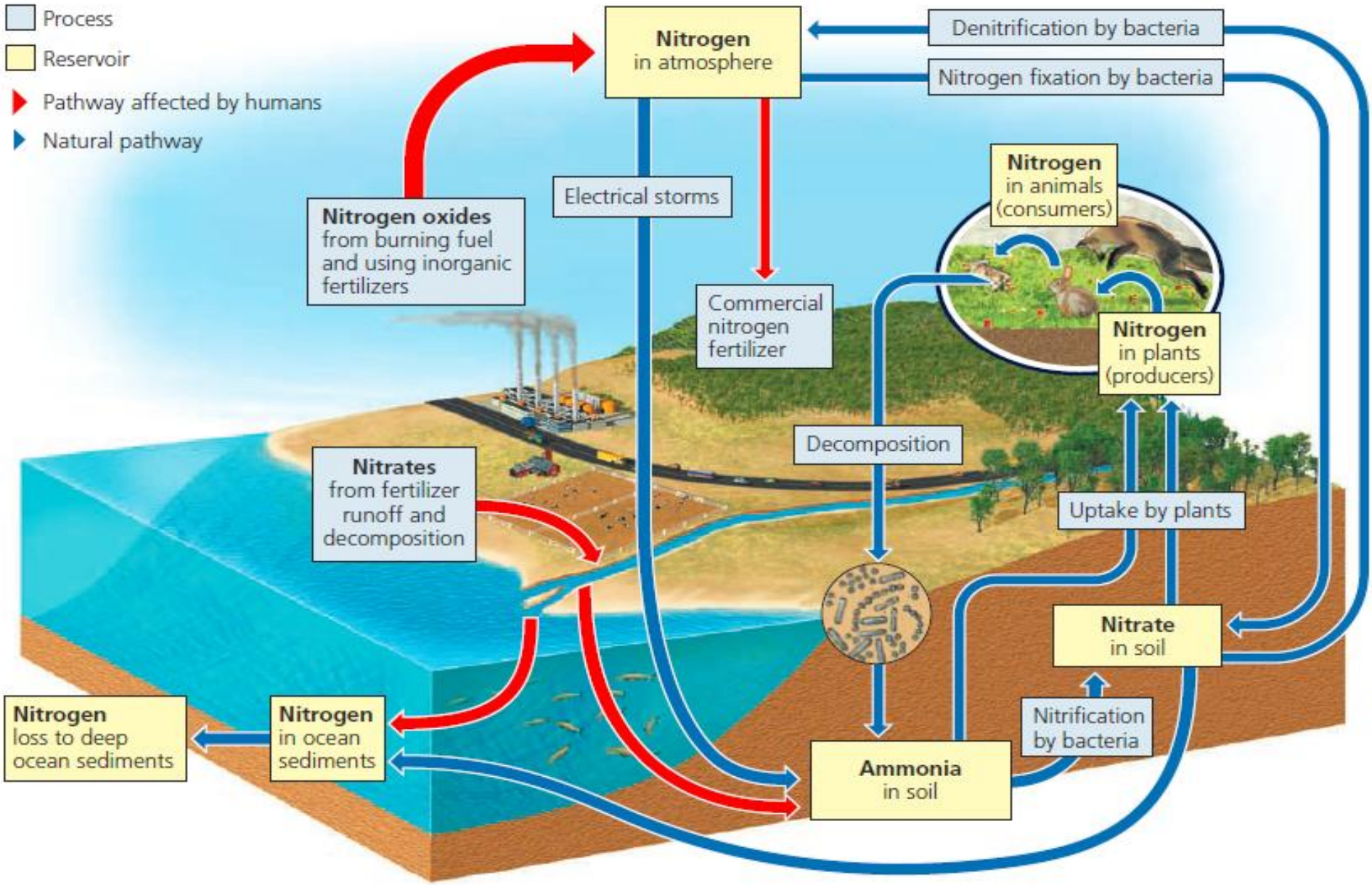
Water Cycle


Condensation
Condensation
Ice and snow
Transpiration from plants
Precipitation to land
Evaporation of surface water
Evaporation from ocean
Runoff
Lakes and reservoirs
Precipitation to ocean
Runoff
Increased runoff on land covered with crops, buildings, and pavement
Infiltration and percolation into aquifer
Increased runoff from cutting forests and filling wetlands
Runoff
Groundwater in aquifers
Overpumping of aquifers
Water pollution
Runoff
Ocean
Natural process
Natural reservoir
Human impacts
Natural pathway
Pathway affected by human activities

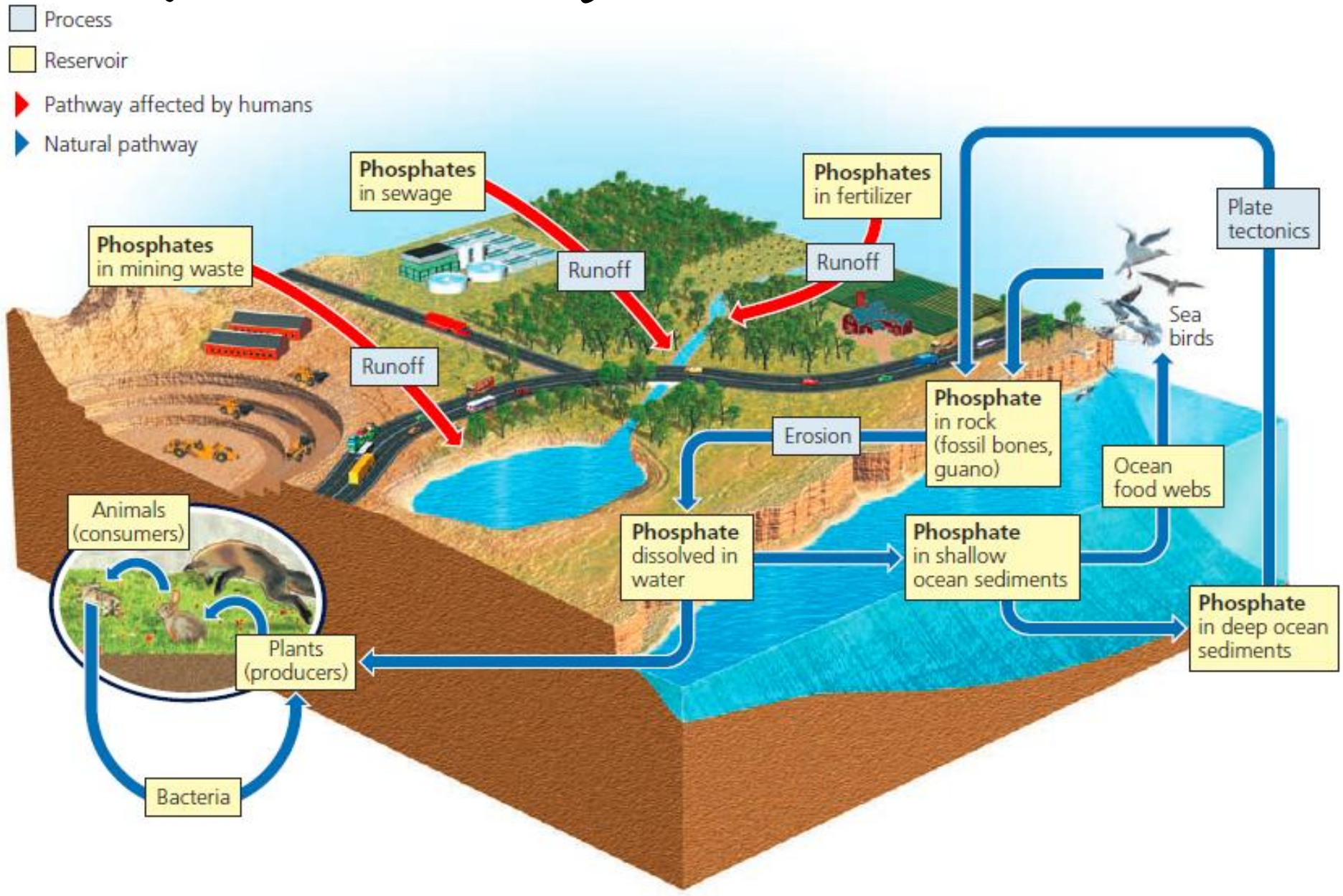
Carbon cycle



Nitrogen Cycle



Phosphorous Cycle



Diversity in classroom





Biodiversity

- **Biodiversity** or **biological diversity** is the variation of taxonomic life forms within a given ecosystem, biome or for the entire Earth.

- **Biodiversity** refers to variety and variability among all groups of living organisms and the ecosystem complexes in which they occur.



Types of Biodiversity

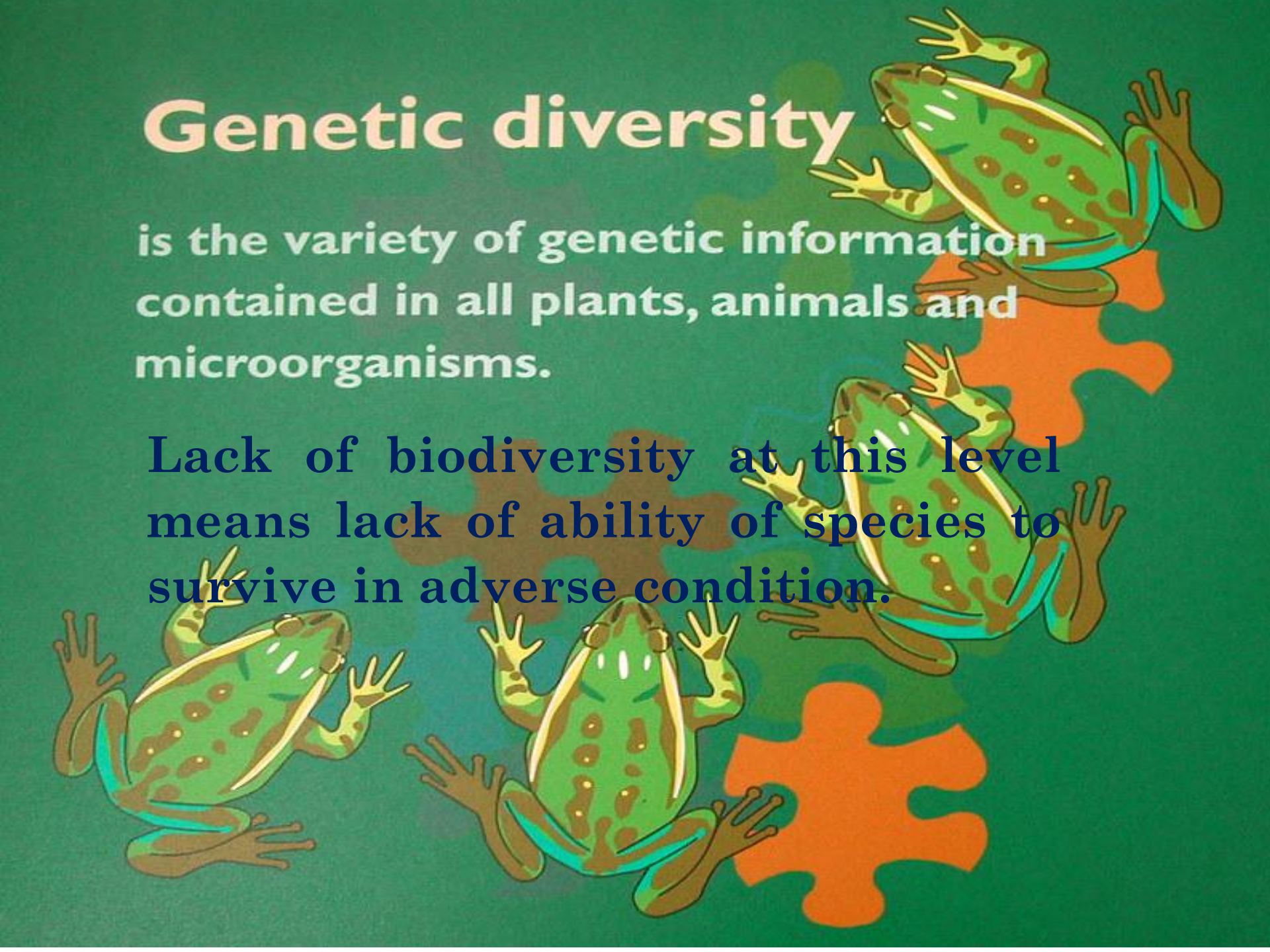
Functional Diversity
The biological and chemical processes such as energy flow and matter recycling needed for the survival of species, communities, and ecosystems.

Ecological Diversity
The variety of terrestrial and aquatic ecosystems found in an area or on the earth.



Genetic Diversity
The variety of genetic material within a species or a population.

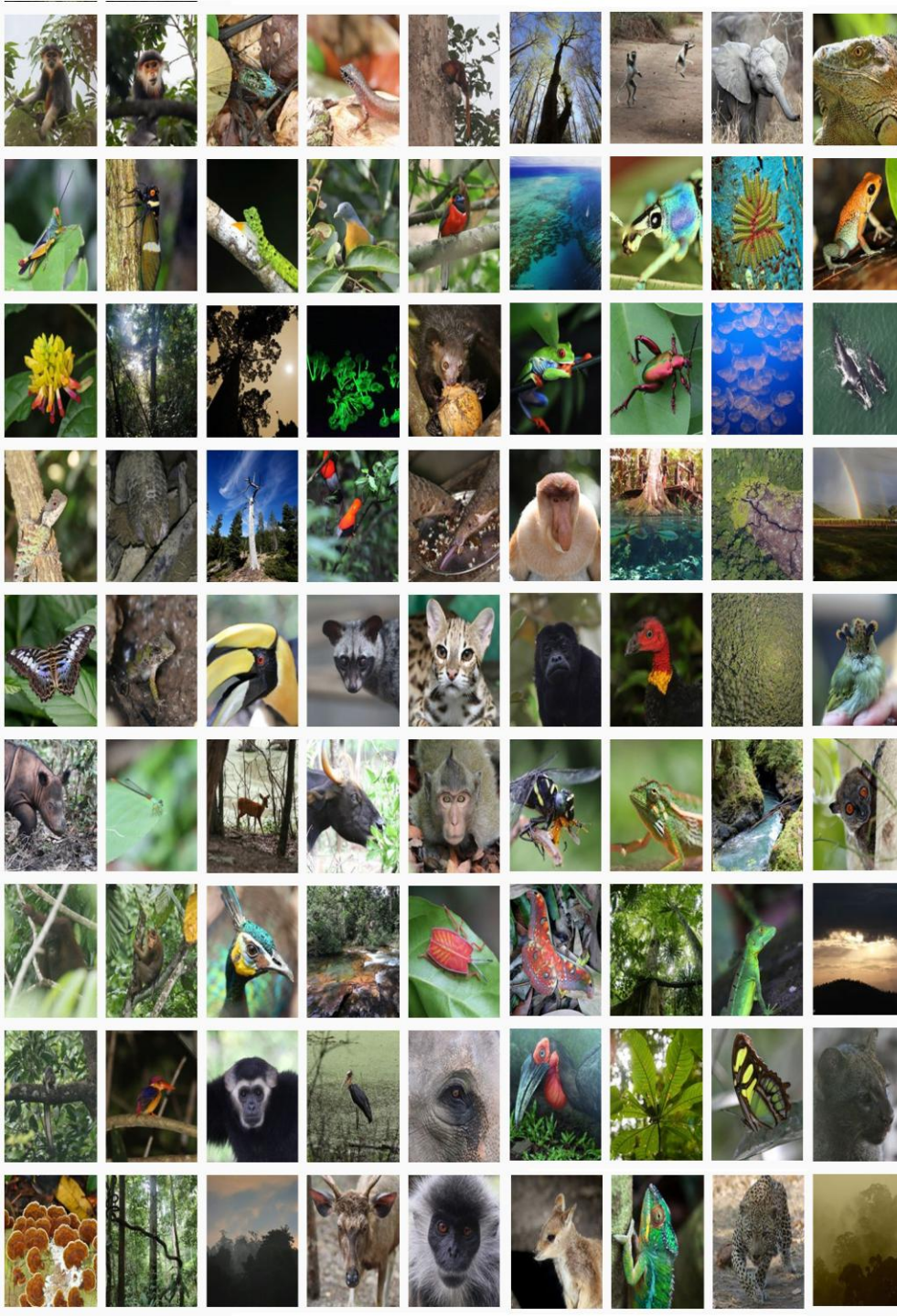
Species Diversity
The number and abundance of species present in different communities.


Genetic diversity
is the variety of genetic information contained in all plants, animals and microorganisms.
Lack of biodiversity at this level means lack of ability of species to survive in adverse condition.

Species Biodiversity

- **Species Diversity is simply the number and relative abundance of species found in a given biological organisation (population, ecosystem, Earth).**

- It varies place to place and even seasonally at the same place. For e.g. different kinds of plants, trees, birds, insects, animals, fishes etc. in a particular location / in an ecosystem.



Simpson's Biodiversity index

Consider the following two ecosystems.

Ecosystem I: 5 Rabbits, 10 frogs and 5 tigers.

Ecosystem II: 3 Rabbits, 4 frogs, 5 tigers and 5 elephants

It is assumed that all rabbits are from the same species. Similarly, also for other animals written here.

Simpson's Biodiversity index (D)

$$D = \frac{\Sigma n(n-1)}{N(N-1)}$$

where, n = the total number of individuals of each species, N = the total number of organisms of all species.

Question

Consider two hypothetical ecosystems: 1 and 2. Ecosystem 1 has four elephants, six tigers, and ten rabbits, whereas Ecosystem 2 has twenty elephants and twenty rabbits. It is assumed that all the elephants belong to one elephant species. The same is assumed true for all the other respective animals. With the help of Simpson's biodiversity index, calculate the same for both the ecosystems. Which ecosystem has more biodiversity as per the above calculation?

Ecosystem Biodiversity

- **This is the diversity of ecological complexity showing variations in physical characters, ecological niches, trophic structure, food-webs, nutrient cycling etc.**
- **Ecosystem Biodiversity is the distinctive group of species that live together in the same area and interact with the physical environment in unique ways.**
- **Different ecosystems (aquatic & terrestrial) are formed due to different geological and environmental conditions prevailing on earth like moisture, temperature, precipitation, altitude etc.**





Identify the type of biodiversity



Identify the type of biodiversity





Identify the type of biodiversity















WHY IS BIODIVERSITY IMPORTANT?

All species, including humans, depend on each other to survive. Loss of biodiversity leads to the loss of services that nature provides that are essential to the functioning of our society and economy.

We depend on biodiverse ecosystems for:



Materials such as food, fuels and fibres



Regulating climate, wastes and pollination

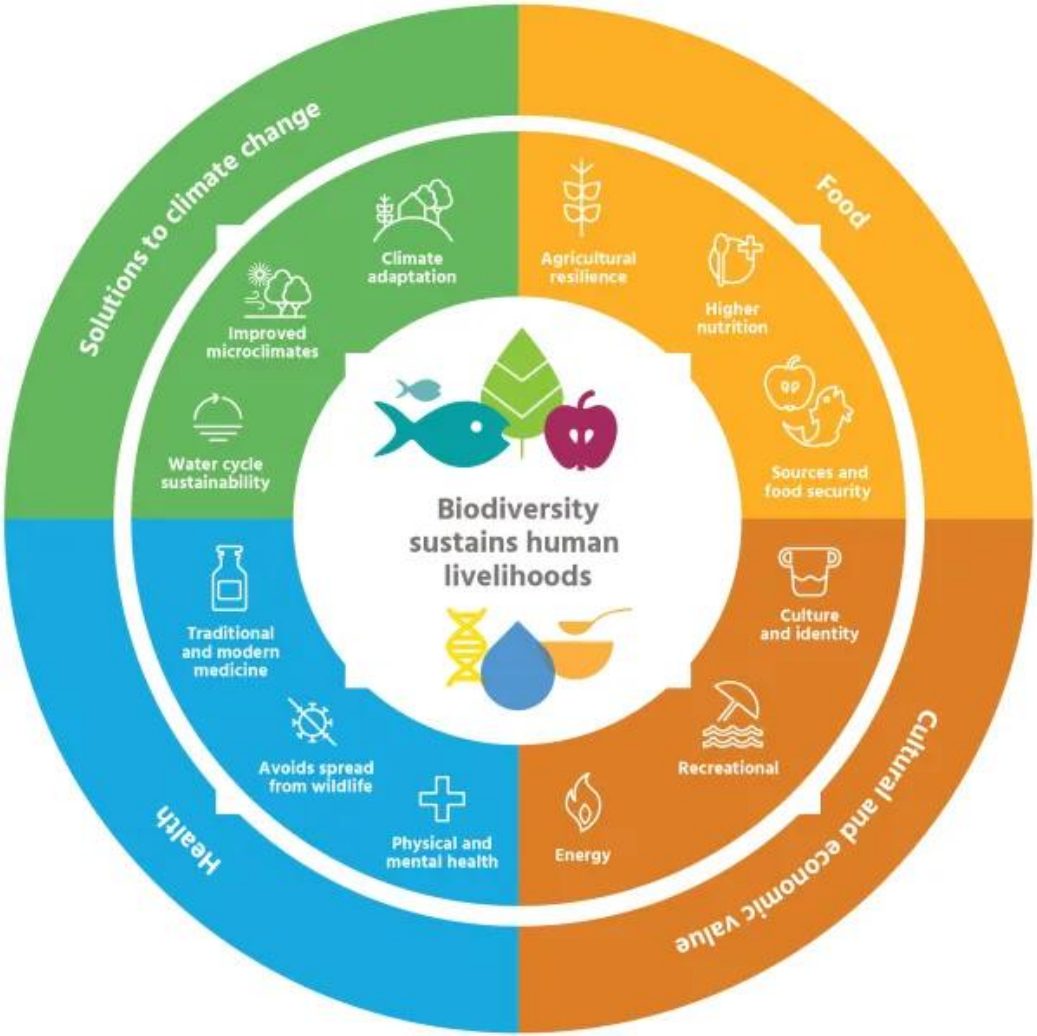


Supporting processes such as water purification and nutrient cycling



Opportunities for enjoyment of the beautiful outdoors

Importance of Biodiversity





1. CLIMATE CHANGE

Increase in the temperature of the atmosphere has major effects on the environment such as the seasons, rising of the sea levels, and glacial retreats.



2. HABITAT LOSS & DEGRADATION

Habitat loss may either be caused by natural events like natural calamities and geological events or anthropogenic activities like deforestation and man-induced climate change.



3. POLLUTION

Be it water, air, or land pollution, all forms of pollution appear to be a threat to all life forms on Earth.



4. INVASIVE SPECIES

An exotic or unnatural species can be any kind of organism that has been introduced to a foreign habitat. This introduction can cause major threats to the native species.



5. OVEREXPLOITATION

Overexploitation refers to the act of over-harvesting species and natural resources at rates faster than they can actually sustain themselves in the wild.



6. OTHER POTENTIAL THREATS

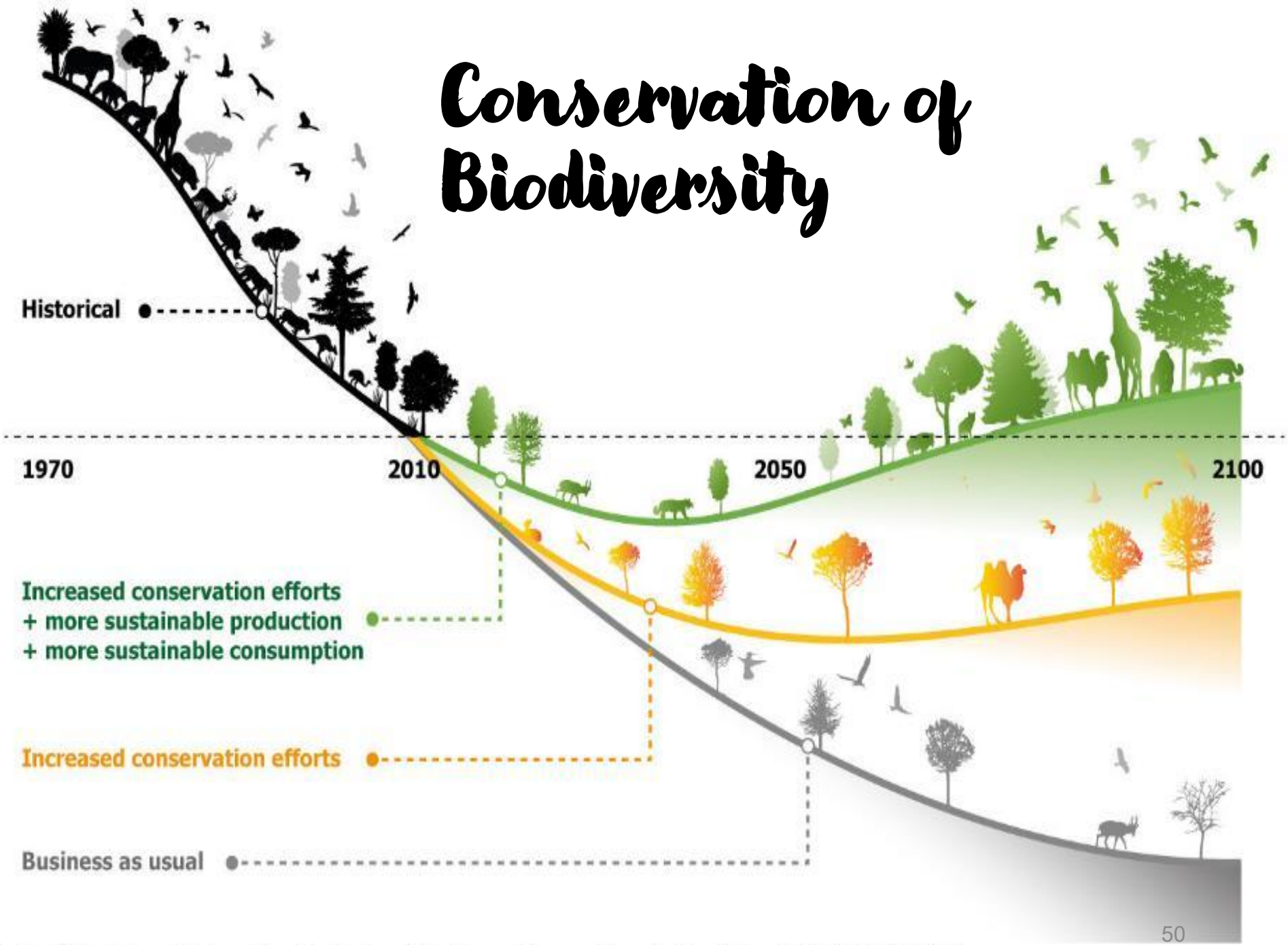
Epidemics and infectious diseases of wildlife such as Ebola virus disease, infectious bursal disease, and flu affect wildlife and biodiversity.

Threats to Biodiversity

"**HIPPO**"

- **H**abitat Loss
- **I**nvasive Species
- **P**oaching
- **P**ollution
- **O**ver population

Conservation of Biodiversity

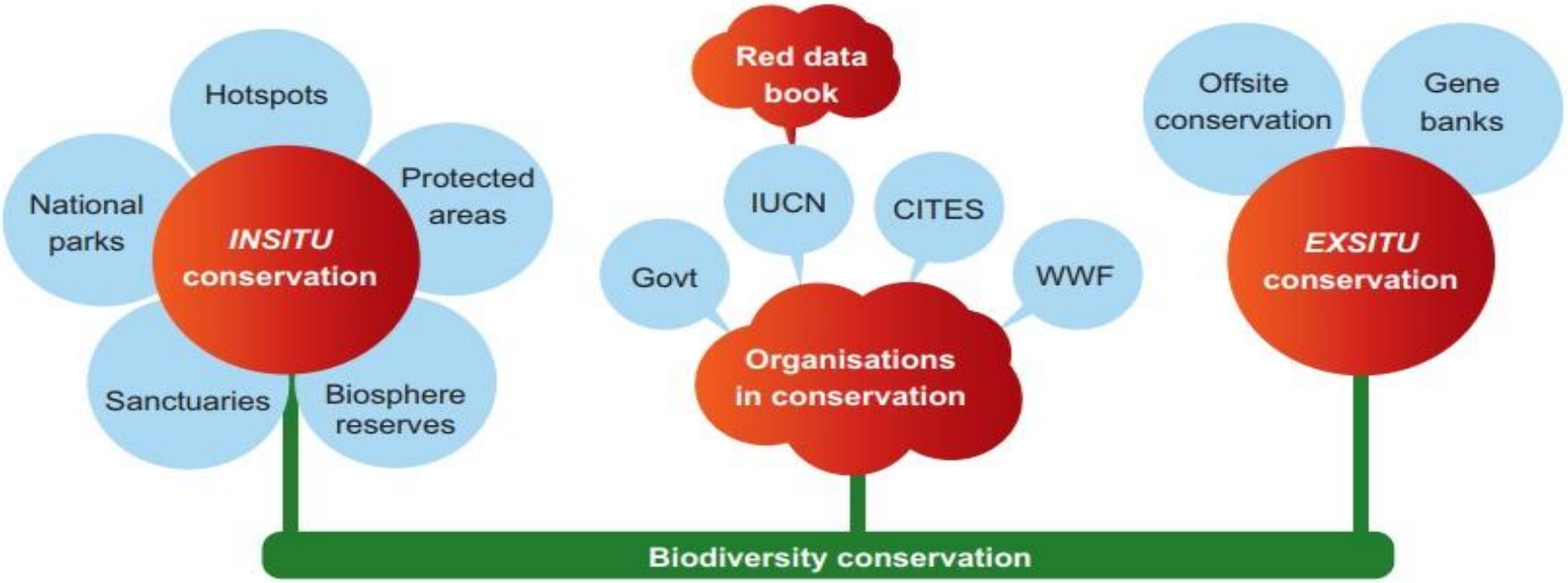


This artwork illustrates the main findings of the article, but does not intend to accurately represent its results (https://doi.org/10.1038/s41586-020-2705-y)

Conservation of Biodiversity

Wildlife life conservation in protected habitat can be done by following approaches:

1) **In-situ conservation**
2) **Ex-situ conservation**



In-situ Conservation of Biodiversity

- In-situ conservation of biodiversity is the conservation of species within their natural habitat. In this method, the natural ecosystem is maintained and protected.
- Conserving a species **in its own environment** by creating national parks or wildlife sanctuaries.

The in-situ conservation has several advantages. Following are the important advantages of in-situ conservation:

1. It is a cost-effective and a convenient method of conserving biodiversity.

2. A large number of living organisms can be conserved simultaneously.

3. Since the organisms are in a natural ecosystem, they can evolve better and can easily adjust to different environmental conditions.

Protected Areas

Protected areas or **conservation areas** are locations which receive protection because of their recognized natural, ecological or cultural values.

- Gir Sanctuary: Asiatic Lion
- Western Ghat: Diverse forest
- Bhimasankar: Rich flora
- Nilgiri Hills: Indian Elephant
- Chilika Lake: Coastal Ecosystem
- Sunderban: Mangrove forest



Ex-situ Conservation of Biodiversity

- Ex-situ or off-site conservation means conservation of species particularly endangered species away from their natural habitat under human supervision. There is less competition for food, water and space among the organisms.
- Many rare species or species having small remaining population are protected by this strategy.
- For e.g.- Zoo, Aquarium, Botanical garden, Seed Bank, Gene Bank







Seed



Gene Bank



Botanical Garden



Zoo

Ex-situ Conservation of Biodiversity

- Ex-situ conservation has the following advantages:

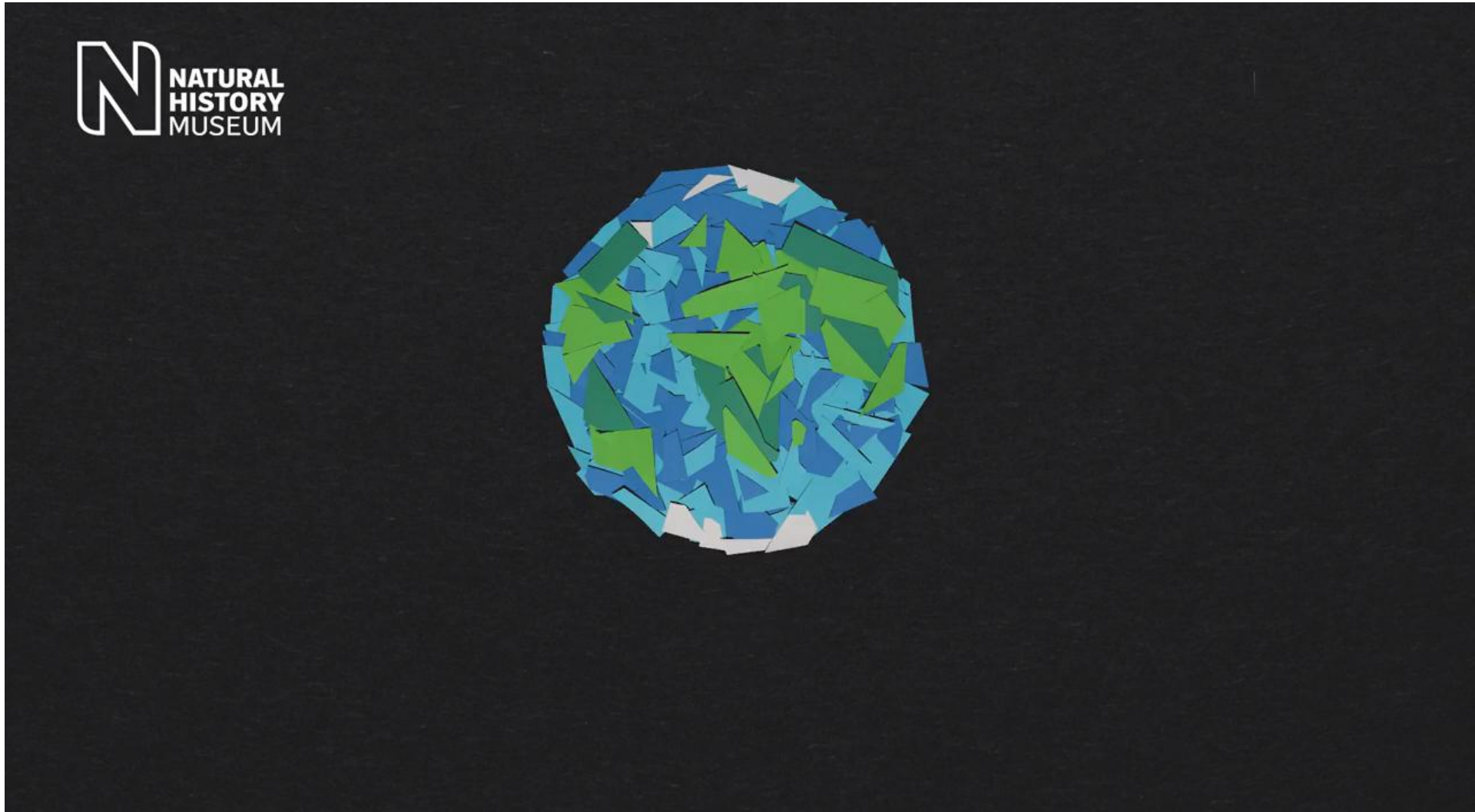
- The animals are provided with a longer time and breeding activity.
- The species bred in captivity can be reintroduced in the wild.
- Genetic techniques can be used for the preservation of endangered species.







What is biodiversity?



What is biodiversity?



Why is biodiversity so important?


Thank you!
Gatis
Sluka